

Internet of Things

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Variables

```
1. int num = 25;
2. float pi = 3.14;
3.
   void setup() {
4.   // put your setup code here, to run once:
5.   Serial.begin(9600);
6. }
7.
   void loop() {
8.   // put your main code here, to run repeatedly:
9.   Serial.print("Hello Welcome \n");
10.  Serial.println("This is Internet of things");
11.  Serial.println("The number is " + String(num) + " The Value of PI is " +
    String(pi));
12.  //Delay
13.  delay(1000);
14. }
```

Using For Loop

```
1.  int num = 25;
2.  float pi = 3.14;
3.  int i = 1;
4.  void setup() {
5.    // put your setup code here, to run once:
6.    Serial.begin(9600);
7.  }
8.
9.  void loop() {
10.   // put your main code here, to run repeatedly:
11.   for (i= 1; i <= 2; i++)
12.     Serial.print("Hello Welcome User Number " + String(i) + "\n");
13.     Serial.println("This is Internet of things");
14.     Serial.println("The number is " + String(num) + " The Value of PI is " + String(pi));
15.     //Delay
16.     delay(1000);
17.  }
```

DEMO 1: USING SERIAL PLOTTER

```
1. void setup() {  
2.   Serial.begin(9600);  
3. }  
  
4. void loop() {  
5.   for (float j = 0; j < 360.00; j=j+2) {  
6.     Serial.println(sin(j * (PI / 180.00))); //plot sinewave  
7.   }  
8. }
```

DEMO 2: Sine & Cosine Plotting


```
1. void setup() {  
2.   Serial.begin(9600);  
3. }  
  
4. void loop() {  
5.   for (float j = 0; j < 360.00; j=j+2) {  
6.     Serial.print(sin(j * (PI / 180.00))); //plot sinewave  
7.     Serial.print(" ");  
8.     Serial.println(cos(j * (PI / 180.00))); //plot cosinewave  
9.   }  
10.}
```

DEMO 3: RANDOM NUMBER GENERATION

```
1. int myvalue = 0;  
2. void setup() {  
3. Serial.begin(9600);  
4. }  
  
5. void loop() {  
6. myvalue = random(100); // Generates between Zero to 99  
7. Serial.print("Hello the random number is ");  
8. Serial.println(myvalue);  
9. delay(1000);  
10. }
```

DIY: MAKE A DICE

DEMO 4: TO INTERFACE LED WITH ARDUINO

DEVICES REQUIRED

- Arduino Board.
- Jumper Cables.
- A to B Cable.
- LED Module.

LED CODE

```
int red = 11;
int green = 10;
int blue = 9;

void setup() {
    // put your setup code here, to run once:
    pinMode(red, OUTPUT);
    pinMode(green, OUTPUT);
    pinMode(blue, OUTPUT);
}

void loop() {
    // put your main code here, to run repeatedly:
    digitalWrite(blue, HIGH);
}
```

Running Python in Arduino

- We need the PyFirmata library to run Python Code on an Arduino Board.
- Step 1: Is to prepare our board by flashing a firmware on it.
- Step 2: Upload **StandardFirmata** on your Arduino Board. It is found under the ***Examples/Firmata***
- Step 3: Open Python Environment (**iotlab/Scripts/activate**) and install the PyFirmata2 Library (**pip install pyfirmata2**).

Blink Code using Python

1. `import pyfirmata2`
2. `import time`
3. `board = pyfirmata2.Arduino('COM8')`
4. `while True:`
5. `board.digital[13].write(1) # ALWAYS CHECK THE SPACE`
6. `time.sleep(1) #Indentations are required in Python`
7. `board.digital[13].write(0)`
8. `time.sleep(1)`

PWM Code using PyFirmata

```
1. import pyfirmata2
2. import time
3. if __name__ == '__main__':
4.     board = pyfirmata2.Arduino('COM8')
5.     print("Communication Started")

6.     LED = board.digital[9]
7.     LED.mode = pyfirmata2.PWM
8.     pwm_counter = 0.01
9.     increase_pwm = True

10.    while True:
11.        if increase_pwm:
12.            pwm_counter += 0.01
13.            if pwm_counter >= 1:
14.                increase_pwm = False
15.        else:
16.            pwm_counter -= 0.01
17.            if pwm_counter <= 0:
18.                increase_pwm = True
19.            LED.write(pwm_counter)
20.            time.sleep(0.01)
```

Assignment

1. WAP to plot the graphs for Sec and Cosec.
2. WAP to plot the graph for a square wave.
3. WAP to Create five new colours.
4. WAP to create a function that can produce any colour.
5. WAP to create a dimming module for the LED.

Please write the same in your Lab Record Books with the Circuit Diagram on the blank page, and upload your codes to Teams as .ino for each assignment. Please sign each code with your name and roll number in comments.

Thanks